PART III: ENTROPY AND THE SECOND LAW OF THERMODYNAMICS

Exercise 1.

If the expansion of the gas is reversible and isothermal, then there's no change in internal energy. However, if the process is reversible and adiabatic, then there would be no change in entropy.

Since the gas is ideal, its pressure p is given in terms of the number of moles n, the volume V, and the temperature T by $p = nRT/V$. If the expansion is isothermal, the work done by the gas is

$$W = \int_{V_1}^{V_2} p\,dV = nRT \int_{V_1}^{V_2} \frac{dV}{V} = nRT \ln \frac{V_2}{V_1},$$

and the corresponding change in entropy is $\Delta S = \int (1/T) dQ = Q/T$, where Q is the heat absorbed.

(a) With $V_2 = 2.00V_1$ and $T = 400$ K, we obtain

$$W = nRT\ln 2.00 = 9.22 \times 10^3 \text{ J}.$$

(b) According to the first law of thermodynamics, $\Delta E_{int} = Q - W$. Now the internal energy of an ideal gas depends only on the temperature and not on the pressure and volume. Since the expansion is isothermal, $\Delta E_{int} = nC_v\Delta T = 0$ and $Q = W$. Thus,

$$\Delta S = \frac{Q}{T} = \frac{W}{T} = 23.1 \text{ J/K}.$$

(c) The change in entropy $\Delta S = 0$ for all reversible adiabatic processes.

Exercise 2.

An isothermal process is one in which $T_i = T_f$, which implies $\ln (T_f / T_i) = 0$. Therefore,

$$\Delta S = nR \ln\left(\frac{V_f}{V_i}\right) \Rightarrow n = \frac{22.0}{(8.31)\ln(3.4/1.3)} = 2.75 \text{ mol}.$$

Exercise 3.

For a reversible isothermal process, we have: $\Delta S = S_f - S_i = \frac{Q}{T}$.

Therefore, $Q = T\Delta S = 1.86 \times 10^4$ J.

Exercise 4.

(a) The energy absorbed as heat is found as $Q = cm\Delta T = 5.79 \times 10^4$ J,

where we have used the fact that a change in Kelvin temperature is equivalent to a change in Celsius degrees.

(b) We have the equation for the entropy change of two blocks coming to equilibrium: $\Delta S = \int (1/T) dQ = mc\ln (T_f/T_i)$.

With $T_f = 373.15$ K and $T_i = 298.15$ K, we obtain $\Delta S = 173$ J/K.

Exercise 5.

(a) This may be considered a reversible process (as well as isothermal), so we use $\Delta S = Q/T$ where $Q = Lm$ with $L = 333$ J/g. Consequently, $\Delta S = 14.6$ J/K.

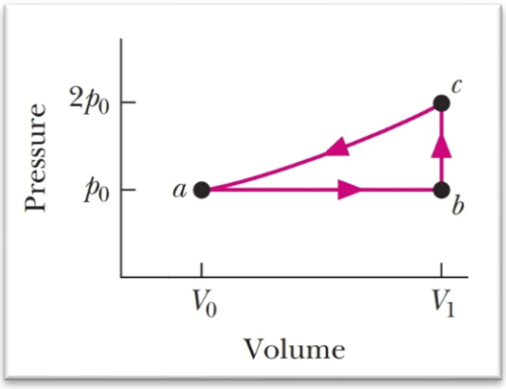
(b) The situation is similar to that described in part (a), except with $L = 2256$ J/g, $m = 5.00$ g, and $T = 373$ K. We therefore find $\Delta S = 30.2$ J/K.

Exercise 6.

We have the equation for the entropy change of two blocks coming to equilibrium: $\Delta S = mc\ln(T_f/T_i)$.

Then, with ΔS = 50 J/K, T_f = 380 K, T_i = 280 K, and m = 0.364 kg, we obtain c = 4.5×102 J/kg.K.

Exercise 7.



(a) Work is done only for the *ab* portion of the process. This portion is at constant pressure, so the work done by the gas is

$$W = \int_{V_0}^{4V_0} p_0\,dV = p_0(4.00V_0 - 1.00V_0) = 3.00p_0V_0 \;\Rightarrow\; \frac{W}{p_0V} = 3.00.$$

(b) We use the first law: $\Delta E_{int} = Q - W$. Since the process is at constant volume, the work done by the gas is zero and $\Delta E_{int} = Q$. The energy Q absorbed by the gas as heat is $Q = nC_V\Delta T$, where C_V is the molar specific heat at constant volume and ΔT is the change in temperature. Since the gas is a monatomic ideal gas, $C_V = 3R/2$. Use the ideal gas law to find that the initial temperature is

$$T_b = \frac{p_bV_b}{nR} = \frac{4p_0V_0}{nR}$$

and that the final temperature is

$$T_c = \frac{p_cV_c}{nR} = \frac{(2p_0)(4V_0)}{nR} = \frac{8p_0V_0}{nR}.$$

Thus, $Q = 6.00p_0V_0$.

The change in the internal energy is $\Delta E_{int} = 6p_0V_0$ or $\Delta E_{int}/p_0V_0 = 6.00$. Since n = 1 mol, this can also be written $Q = 6.00RT_0$.

(c) For a complete cycle, $\Delta E_{int} = 0$.

(d) Since the process is at constant volume, use $dQ = nC_VdT$ to obtain

$$\Delta S = \int \frac{dQ}{T} = nC_V\int_{T_b}^{T_c}\frac{dT}{T} = nC_V\ln\frac{T_c}{T_b}.$$

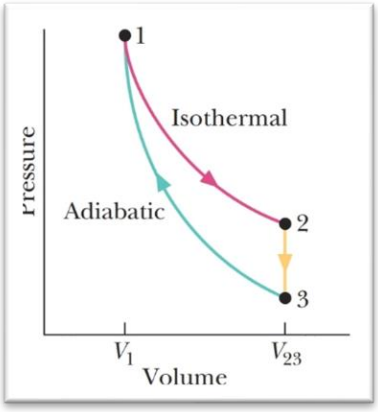
Substituting $C_V = 3R/2$ (monatomic gas) and using the ideal gas law, we write

$$\frac{T_c}{T_b} = \frac{p_cV_c}{p_bV_b} = \frac{(2p_0)(4V_0)}{p_0(4V_0)} = 2.$$

Thus, $\Delta S = 3/2nR\ln 2$. Since n = 1, this is ΔS = 8.64 J/K.

(e) For a complete cycle, $\Delta E_{int} = 0$ and $\Delta S = 0$.

Exercise 8.



The connection between molar heat capacity and the degrees of freedom of a diatomic gas is given by setting $f = 5$. Thus, $C_V = (f/2)R = 5R/2$, $C_P = (f/2)R + R = 7R/2$, and $\gamma = 7/5$. We note that we are asked to use the ideal gas constant as R and not plug in its numerical value. We also recall that isothermal means constant temperature, so $T_2 = T_1$ for the 1→2 process. The statement (at the end of the problem) regarding "per mole" may be taken to mean that n may be set identically equal to 1 wherever it appears.

(a) The gas law in ratio form is used to obtain

$$p_2 = p_1\left(\frac{V_1}{V_2}\right) = \frac{p_1}{3} \quad \Rightarrow \quad \frac{p_2}{p_1} = \frac{1}{3} = 0.333\,.$$

(b) The adiabatic relations lead to

$$p_3 = p_1\left(\frac{V_1}{V_3}\right)^{\gamma} = \frac{p_1}{3^{1.4}} \quad \Rightarrow \quad \frac{p_3}{p_1} = \frac{1}{3^{1.4}} = 0.215\,.$$

(c) Similarly, we find

$$T_3 = T_1\left(\frac{V_1}{V_3}\right)^{\gamma-1} = \frac{T_1}{3^{0.4}} \quad \Rightarrow \quad \frac{T_3}{T_1} = \frac{1}{3^{0.4}} = 0.644.$$

(process 1→2).

(d) The work is given as $W = nRT_1\ln(V_2/V_1) = RT_1\ln 3 = 1.10RT_1$. Thus, $W/nRT_1 = \ln 3 = 1.10$.

(e) The internal energy change is $\Delta E_{int} = 0$, since this is an ideal gas process without a temperature change. Thus, the energy absorbed as heat is given by the first law of thermodynamics: $Q = \Delta E_{int} + W \approx 1.10RT_1$, or $Q/nRT_1 = \ln 3 = 1.10$.

(f) $\Delta E_{int} = 0$ or $\Delta E_{int}/nRT_1 = 0$.

(g) The entropy change is $\Delta S = Q/T_1 = 1.10R$, or $\Delta S/R = 1.10$

(process 2→3).

(h) The W = 0, since there is no volume change. Therefore, $W/nRT_1 = 0$.

(i) The internal energy change is

$$\Delta E_{int} = nC_V\left(T_3 - T_2\right) = (1)\left(\frac{5}{2}R\right)\left(\frac{T_1}{3^{0.4}} - T_1\right) \approx -0.889\,RT_1 \quad \Rightarrow \quad \frac{\Delta E_{int}}{nRT_1} \approx -0.889.$$

This ratio (-0.889) is also the value for Q/nRT_1 (by either the first law of thermodynamics or by the definition of C_V).

(j) $\Delta E_{int}/nRT_1 = -0.889$.

(k) For the entropy change, we obtain

$$\Delta S/R = n\ln(V3/V2) + nCv\ln(T3/T2)/R = \ln(1) + 5/2\ln[(T_1/3^{0.4})/T_1] = -1.1$$

(process 3→1).

(l) By definition, $Q = 0$ in an adiabatic process, which also implies an absence of entropy change (taking this to be a reversible process). The internal change must be the negative of the value obtained for it in the previous process (since all the internal energy changes must add up to zero, for an entire cycle, and its change is zero for process 1→2), so $\Delta E_{int} = +0.889RT_1$. By the first law of thermodynamics, then, $W = Q - \Delta E_{int} = -0.889RT_1$, or $W/nRT_1 = -0.889$.

(m) $Q = 0$ in an adiabatic process.

(n) $\Delta E_{int}/nRT_1 = 0.889$.

(o) $\Delta S/nR = 0$.

Exercise 9.

We note that the connection between molar heat capacity and the degrees of freedom of a monatomic gas is given by setting $f = 3$. Thus, $C_V = 3R/2$, $C_P = 5R/2$, and $\gamma = 5/3$.

(a) Since this is an ideal gas, $\Delta E_{int} = 0$ for this process. By the first law of thermodynamics, $Q = 0 + W = nRT_1\ln V_2/V_1 = p_1V_1\ln 2 \rightarrow Q/p_1V_1 = \ln 2 = 0.693$.

(b) The gas law in ratio form implies that the pressure decreased by a factor of 2 during the isothermal expansion process to $V_2 = 2.00V_1$, so that it needs to increase by a factor of 4 in this step in order to reach a final pressure of $p_2 = 2.00p_1$. That same ratio form now applied to this constant-volume process, yielding $4.00 = T_2T_1$, which is used in the following: $Q = nC_V\Delta T = 9/2p_1V_1$, or $Q/p_1V_1 = 9/2 = 4.50$.

(c) The work done during the isothermal expansion process can be obtained as: $W = nRT_1\ln V_2/V_1 = p_1V_1\ln 2.00 \rightarrow W/p_1V_1 = \ln 2 = 0.693$.

(d) In step 2 where the volume is kept constant, $W = 0$.

e) The change in internal energy can be calculated by combining the above results and applying the first law of thermodynamics: $\Delta E_{int} = Q_{total} - \underline{W}_{total} = 9/2p_1V_1$, or $\Delta E_{int}/\ p_1V_1 = 9/2 = 4.50$.

(f) The change in entropy:

$$\Delta S = R\ln\left(\frac{2.00V_1}{V_1}\right) + C_V\ln\left(\frac{4.00T_1}{T_1}\right) = R\ln 2.00 + \left(\frac{3}{2}R\right)\ln(2.00)^2$$
$$= R\ln 2.00 + 3R\ln 2.00 = 4R\ln 2.00 = 23.0\ \text{J/K}.$$

The second approach consists of an isothermal (constant T) process in which the volume halves, followed by an isobaric (constant p) process.

(g) Here the gas law applied to the first (isothermal) step leads to a volume half as big as the original. Since $\ln(1/2) = -\ln 2$, the reasoning used above leads to: $Q = -p_1V_1\ln 2 \rightarrow Q/p_1V_1 = -\ln 2 = -0.693$.

(h) To obtain a final volume twice as big as the original, in this step we need to increase the volume by a factor of 4.00. Now, the gas law applied to this isobaric portion leads to a temperature ratio $T_2/T_1 = 4$. Thus, $Q = C_P\Delta T = 5/2R(T_2-T_1) = 15/2p_1V_1$, or $Q/p_1V_1 = 15/2 = 7.50$.

(i) During the isothermal compression process, we have: $W = nRT_1\ln V_2/V_1 = p_1V_1\ln(-1/2) = -p_1V_1\ln 2 \rightarrow W/p_1V_1 = -\ln 2 = -0.693$.

(j) The initial value of the volume, for this part of the process, is $V_i = V_1/2$, and the final volume is $V_f = 2V_1$. The pressure maintained during this process is $p' = 2p_1$. The work is: $W = p'\Delta V = p'(V_f - V_i) = 3p_1V_1 \rightarrow W/p_1V_1 = 3$.

(k) Using the first law of thermodynamics, the change in internal energy is: $\Delta E_{int} = Q_{total} - W_{total} = (15/2p_1V_1 - p_1V_1\ln 2) - (3p_1V_1 - p_1V_1\ln 2) = 9/2p_1V_1 \rightarrow \Delta E_{int}/p_1V_1 = 9/2 = 4.50$. The result is the same as that obtained in part (e).

(l) Similarly, $\Delta S = 4R\ln 2 = 23.0$ J/K. the same as that obtained in part (f)

Exercise 10.

With $T_L = 290$ k, we find $\varepsilon = 1 - T_L/T_H \rightarrow T_H = T_L/(1 - \varepsilon) = 483$K.

If we replace $\varepsilon = 0.40$ in the above calculation with $\varepsilon = 0.50$, we obtain a (final) high temperature equal to $T_H' = 580$ K. The difference is $T_H' - T_H = 97$ K.

Exercise 11.

The answers to this exercise do not depend on the engine being of the Carnot design. Any heat engine that intakes energy as heat (from, say, consuming fuel) equal to $|Q_H| = 52$ kJ and exhausts (or discards) energy as heat equal to $|Q_L| = 36$ kJ will have these values of efficiency ε and net work W.

(a) $\varepsilon = 1 - |Q_L/Q_H| = 0.31 = 31\%$.

(b) $W = |Q_H| - |Q_L| = 16$ kJ.

Exercise 12.

We solve part (b) first.

(b) For a Carnot engine, the efficiency is related to the reservoir temperatures. Therefore, $T_H = (T_H - T_L)/\varepsilon = 341$ K = 68 °C.

(a) The temperature of the cold reservoir is $T_L = T_H - 75 = 341$ K – 75 K = 266 K.

Exercise 13.

(a) The net work done is the rectangular "area" enclosed in the pV diagram:

$W = (V - V_0)(p - p_0) = (2V_0 - V_0)(2p_0 - p_0) = V_0p_0$.

Substituting the given values into the equation $\rightarrow W = 2.27$ kJ.

(b) We can compute the energy added as heat during the "heat-intake" portions of the cycle:

$$Q_{abc} = nC_V(T_b - T_a) + nC_p(T_c - T_b) = n\left(\frac{3}{2}R\right)T_a\left(\frac{T_b}{T_a} - 1\right) + n\left(\frac{5}{2}R\right)T_a\left(\frac{T_c}{T_a} - \frac{T_b}{T_a}\right)$$

$$= nRT_a\left(\frac{3}{2}\left(\frac{T_b}{T_a} - 1\right) + \frac{5}{2}\left(\frac{T_c}{T_a} - \frac{T_b}{T_a}\right)\right) = p_0V_0\left(\frac{3}{2}(2-1) + \frac{5}{2}(4-2)\right)$$

$$= \frac{13}{2}p_0V_0$$

where, to obtain the last line, the gas law in ratio form has been used. Therefore, since $W = p_0V_0$, we have $Q_{abc} = 13W/2 = 14.8$ kJ.

(c) The efficiency is calculated as:

$\varepsilon = W/|Q_H| = 2/13 = 0.154 = 15.4\%$

(d) A Carnot engine operating between T_c and T_a has efficiency equal to $\varepsilon = 1 - (T_a/T_c) = 1 - 1/4 = 0.75 = 75\%$, where the gas law in ratio form has been used.

(e) This is greater than our result in part (c), as expected from the second law of thermodynamics.

Exercise 14.

(a) We use $\varepsilon = W/|Q_H|$. The heat absorbed is $|Q_H| = |W|/\varepsilon = 33$ kJ.

(b) The heat exhausted is then $|Q_L| = |Q_H| - |W| = 25$ kJ.

(c) Now we have $|Q_H| = |W|/\varepsilon = 26$ kJ.

(d) Similarly, $|Q_L| = |Q_H| - |W| = 18$ kJ.

Exercise 15.

(a) For process D→A:

$$p_D V_D^{\gamma} = p_A V_A^{\gamma} \quad \Rightarrow \quad \frac{p_0}{32}(8V_0)^{\gamma} = p_0 V_0^{\gamma}$$

which leads to $8^{\gamma} = 32 \rightarrow \gamma = 5/3$ ➔ the gas is monatomic.

(b) The input heat is that absorbed during process $A \rightarrow B$:

$$Q_H = nC_p\Delta T = n\left(\frac{5}{2}R\right)T_A\left(\frac{T_B}{T_A} - 1\right) = nRT_A\left(\frac{5}{2}\right)(2-1) = p_0V_0\left(\frac{5}{2}\right)$$

and the exhaust heat is that liberated during process $C \rightarrow D$:

$$Q_L = nC_p\Delta T = n\left(\frac{5}{2}R\right)T_D\left(1 - \frac{T_L}{T_D}\right) = nRT_D\left(\frac{5}{2}\right)(1-2) = -\frac{1}{4}p_0V_0\left(\frac{5}{2}\right)$$

where in the last step we have used the fact that $T_D = \frac{1}{4}T_A$ (from the gas law in ratio form).

Therefore: $\varepsilon = W/Q_H = 75\%$ (phải tính W của từng đoạn, chú ý ko phải carnot engine)

Exercise 16.

(a) The pressure at 2 is $p_2 = 3p_1$, as given in the problem statement. The volume is $V_2 = V_1 = nRT_1/p_1$. The temperature is $T_2 = p_2V_2/nR = 3T_1 \rightarrow T_2/T_1 = 3$.

(b) The process 2→3 is adiabatic, so $T_2V_2^{\gamma-1} = T_3V_3^{\gamma-1}$. Using the result from part (a), $V_3 = 4V_1$, $V_2 = V_1$, and $\gamma = 1.3$, we obtain $T_3/T_1 = 1.98$.

(c) The process 4→1 is adiabatic, so $T_4V_4^{\gamma-1} = T_1V_1^{\gamma-1}$. Since $V_4 = 4V_1$, we have $T_4/T_1 = 0.66$.

(d) The process 2→3 is adiabatic, so $p_2V_2^{\gamma} = p_3V_3^{1}$ or $p_3 = (V_2/V_3)^{\gamma}p_2$. Substituting $V_3 = 4V_1$, $V_2 = V_1$, $p_2 = 3p_1$, and $\gamma = 1.3$, we obtain $p_3/p_1 = 0.495$.

(e) The process 4→1 is adiabatic, so $p_4V_4^{\gamma} = p_1V_1^{1}$, and $p_4/p_1 = 0.165$.

(f) The efficiency of the cycle is $\varepsilon = W/Q_{12}$, where W is the total work done by the gas during the cycle and Q_{12} is the energy added as heat during the 1→2 portion of the cycle, the only portion in which energy is added as heat. The work done during the portion of the cycle from 2 to 3 is $W_{23} = \int pdV$. Substitute $p = p_2V_2^{\gamma}/V^{\gamma}$ to obtain:

$$W_{23} = p_2V_2^{\gamma}\int_{V_2}^{V_3} V^{-\gamma}dV = \left(\frac{p_2V_2^{\gamma}}{\gamma-1}\right)\left(V_2^{1-\gamma} - V_3^{1-\gamma}\right).$$

Substitute $V_2 = V_1$, $V_3 = 4V_1$, and $p_3 = 3p_1$ to obtain

$$W_{23} = \left(\frac{3p_1V_1}{1-\gamma}\right)\left(1-\frac{1}{4^{\gamma-1}}\right) = \left(\frac{3nRT_1}{\gamma-1}\right)\left(1-\frac{1}{4^{\gamma-1}}\right).$$

Similarly, the work done during the portion of the cycle from 4 to 1 is

$$W_{41} = \left(\frac{p_1V_1^{\gamma}}{\gamma-1}\right)\left(V_4^{1-\gamma} - V_1^{1-\gamma}\right) = -\left(\frac{p_1V_1}{\gamma-1}\right)\left(1-\frac{1}{4^{\gamma-1}}\right) = -\left(\frac{nRT_1}{\gamma-1}\right)\left(1-\frac{1}{4^{\gamma-1}}\right).$$

No work is done during the 1→2 and 3→4 portions, so the total work done by the gas

during the cycle is

$$W = W_{23} + W_{41} = \left(\frac{2nRT_1}{\gamma-1}\right)\left(1-\frac{1}{4^{\gamma-1}}\right).$$

The energy added as heat is $Q_{12} = nC_V(T_2-T_1) = nC_V(3T_1-T_1) = 2nC_VT_1$, where C_V is the molar specific heat at constant volume. Now $\gamma = C_P/C_V = (C_V + R)/C_V = 1 + (R/C_V) \rightarrow C_V = R/(\gamma - 1)$

CP is the molar specific heat at constant pressure, which for an ideal gas is $C_P = C_V + R$. Thus, $Q_{12} = 2nRT_1/(\gamma - 1)$. The efficiency is $\varepsilon = 1 - 1/4^{\gamma-1}$, with $\gamma = 1.3$, the efficiency is $\varepsilon = 0.34 = 34\%$.

Exercise 17.

Considering a Carnot refrigerator, we have:

$|W| = |Q_L|/K_C = |Q_L|/[T_L/(T_H - T_L)] = 1\text{J}/[(273+7)/(27-7)] = 0.071$ J.

(b) A similar calculation (being sure to use absolute temperature) leads to 0.50 J in this case.

(c) With $T_L = 100$ K, we obtain $|W| = 2$ J.

(d) Finally, with the low temperature reservoir at 50 K, an amount of work equal to $|W| = 5.0$ J is required.

Exercise 18.

The performance of the refrigerator is related to its rate of doing work.

The coefficient of performance for a refrigerator is given by $K = |Q_L|/|W|$, where Q_L is the energy absorbed from the cold reservoir as heat and W is the work done during the refrigerator cycle, a negative value. The first law of thermodynamics yields $Q_H + Q_L - W = 0$ for an integer number of cycles. Here Q_H is the energy ejected to the hot reservoir as heat. Thus, $Q_L = W - Q_H$. Since Q_H is negative and greater than W, so $|Q_L| = |Q_H| - |W|$. Thus, $K = (|Q_H| - |W|)/|W|$.

Therefore, $|W| = |Q_H|/(K + 1)$.

In one hour, $|Q_H| = 7.54$ MJ. With $K = 3.8$, the work done is $|W| = 1.57$ MJ.

The rate at which work is done is $P = |W|/\Delta t = 440$ W.

Exercise 19.

(a) Considering a freezer, which is a reverse Carnot cycle, we have:

$K_C = |Q_L|/(|Q_H| - |Q_L|) \rightarrow |Q_H| = |Q_L|(1 + K_C)/K_C = 49$ kJ.

(b) We obtain $|W| = |Q_H| - |Q_L| = 7.4$ kJ

if we take the initial 42 kJ datum to be accurate to three figures. The given temperatures are not used in the calculation; in fact, it is possible that the given room temperature value is not meant to be the high temperature for the (reversed) Carnot cycle — since it does not lead to the given K_C.

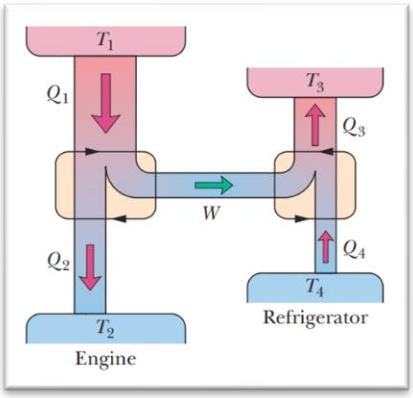
Exercise 20.

The work done by the motor in $t = 10$ min is $|W| = Pt = (200\text{ W})(10.0\text{ min})(60\text{ s/min})$

$= 1.20 \times 10^5$ J. The heat extracted is then

$K = |Q_L|/|W| \rightarrow |Q_L| = K|W| = (T_L|W|)/(T_H - T_L) = 1.08 \times 10^6$ J.

Exercise 21.



The efficiency of the engine is defined by $\varepsilon = W/Q_1$ and is shown in the text to be $\varepsilon = (T_1 - T_2)/T_1 \rightarrow W/Q_1 = (T_1 - T_2)/T_1$.

The coefficient of performance of the refrigerator is defined by $K = Q_4/W$ and is shown in the text to be $K = T_4/(T_3 - T_4) \rightarrow Q_4/W = T_4/(T_3 - T_4)$.

Now $Q_4 = Q_3 - W$, so $(Q_3 - W)/W = T_4/(T_3 - T_4)$.

The work done by the engine is used to drive the refrigerator, so W is the same for the two. Solve the engine equation for W and substitute the resulting expression into the refrigerator equation. The engine equation yields $W = (T_1 - T_2)Q_1/T_1$ and the substitution yields:

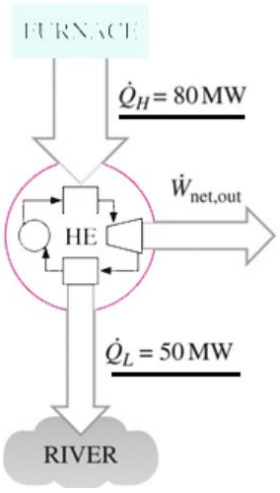
$T_4/(T_3 - T_4) = (Q_3 - W) - 1 = \{(Q_3 T_1)/[Q_1(T_1 - T_2)]\} - 1$.

Solving for Q_3/Q_1, we obtain: $Q_3/Q_1 = [1 - (T_2/T_1)]/[1 - (T_4/T_3)]$.

Substituting $T_1 = 400$ K, $T_2 = 150$ K, $T_3 = 325$ K, and $T_4 = 225$ K, the ratio becomes $Q_3/Q_1 = 2.03$.

MCQs :

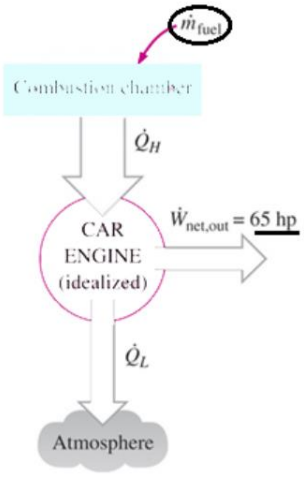
1. Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output and thermal efficiency for this heat engine:



a) 30 MW & 37.5 %
b) 40 MW & 0.375 %
c) 50 MW & 35.7 %
d) 60 MW & 0.357 %

Answer: (a) Net power output = 80 – 50 MW = 30 MW. The thermal efficiency of heat engine = net work output / heat input = 30/80 = 0.375 = 37.5 %

2. A car engine with a power output of 50 kW has a thermal efficiency of 24 percent. Determine the fuel consumption rate of this car if the fuel has a heating value of 44,000 kJ/kg



a) 0.00273 kg/s
b) 0.00373 kg/s
c) 0.00473 kg/s
d) 0.00573 kg/s

Answer: c

Explanation: Q = 50/0.24 = 208.3 kW,

hence fuel consumption rate = 208.3 kW / 44000 kJ/kg = 0.00473 kg/s.

3. A machine which can do work by making use of heat that flows out spontaneously from a high temperature source to a low temperature sink is called:

a. Carnot engine

b. Cyclic machine

c. Heat engine

d. Heat machine

Answer: (c)

4. What is true about entropy?

a. if it increases in one place, it must decrease elsewhere

b. If it decreases in one place, it must increase elsewhere

c. It must decrease in the universe

d. It must stay constant in the universe

Answer: (b)

Entropy is a measure of the disorder or randomness in a system. According to the second law of thermodynamics, the total entropy of an isolated system always increases over time. Therefore, if entropy decreases in one place, it must increase elsewhere in order to maintain the overall increase in entropy in the system. This is because the decrease in entropy in one area would result in a corresponding increase in order or organization, which must be compensated by an increase in disorder or randomness somewhere else.

5. Efficiency of a heat engine is the ratio of:

a. Heat taken from the high temperature reservoir (Q) to the work obtained in a cyclic process (W)

b. Work obtained in a cyclic process (W) to the heat taken from the high temperature reservoir (Q_H)

c. Work obtained in a cyclic process (W) to the heat taken from the low temperature sink (Q_L)

d. Heat taken from the low temperature sink (Q_L) to the work obtained in a cyclic process (W)

Answer: (b)

Efficiency of a heat engine = W/Q_H.

6. In a heat engine, what is the relationship between Q_H, Q_L and W?

a. $W = Q_H + Q_L$

b. $W = Q_L - Q_H$

c. $W = Q_H - Q_L$

d. $W + Q_H + Q_L = 0$

Answer: (c) $W = Q_H - Q_L$

Q_H is turned into work and some heat is ejected

Therefore $Q_H = W + Q_L$

Hence $W = Q_H - Q_L$

7. An air conditioner provides 1 kg/s of air at 15 oC cooled from outside atmospheric air at 35 oC. Estimate the amount of power needed to operate the air-conditioner:

(Cp of air = 1.005 kJ/kgK)

a. 1.09 kW

b. 1.19 kW

c. 1.29 kW

d. 1.39 kW

Answer: (d) 1.39 kW

$Q = mC_p\Delta T = 20.08$ kW

COP = (15 + 273)/(35-15) = 14.4

=> Power needed = 20/14.4 = 1.39 kW

8. A car engine operates with a thermal efficiency of 35 %. Assuming the air-conditioner has a coefficient of performance (COP) of 3 working as a refrigerator cooling the inside using engine shaft work to drive it. How much extra fuel needed to remove 1 kJ from the inside of the car?

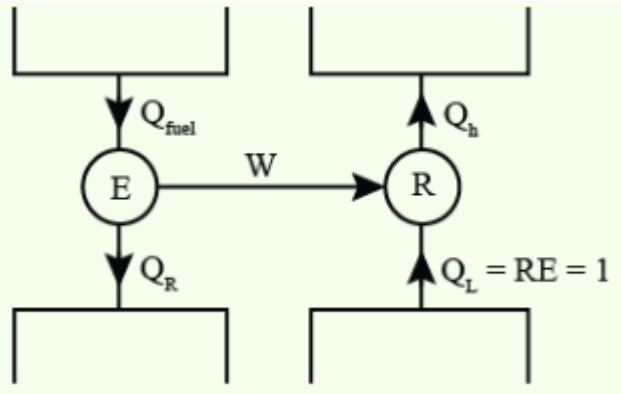
a. 0.752 kJ

b. 0.952 kJ

c. 0.852 kJ

d. None of the above

Answer: (b) 0.952 kJ



For air-conditioner

COP = 3 = RE/W_{in}

=> W = 1/3

For engine

$\eta_c = W/Q_{fuel}$

=> $Q_{fuel} = W/\eta_c = \frac{1}{3}/0.35 = 0.952$ kJ

9. A cyclic machine receives 325 kJ from a 1000 K energy reservoir. It rejects 125 kJ to a 400 K reservoir and the cycle produces 200 kJ of work as output. This cycle is:

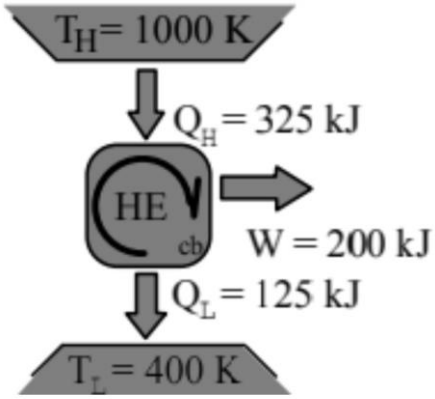
a. reversible

b. irreversible

c. impossible

d. None of the above

Answer: (c) Impossible



The Carnot efficiency = 1 – (400/1000) = 0.6 and the real efficiency = (200/325) = 0.615 which is greater than Carnot efficiency => impossible.

10. The increase in entropy when one mole of ice melts to form water is: (Latent heat of fusion of ice is 80 cal/g)

a. 0.293 Cal/molK

b. 0.65 Cal/molK

c. 5.275 Cal/molK

d. 3.41 Cal/molK

Answer: (c) Q for 1 mole of ice = 80 x 18 = 1440 cal/mol; T = 273 K; => $\Delta S = q/T = 1440/273 = 5.275\ \mathrm{Cal/Kmol}$